

Lab7handout

Lab Assignment 7 : Processing declarations within a single scope, symbol table creation and relevant semantic analysis.

Grammar G1	Grammar G2
R1 : dlist $\rightarrow$ dlist D; { } R2 : dlist $\rightarrow$ D; { } R3: D $\rightarrow$ T id {D.type=T.type; D.size = T.size; print(id.name, D.type, D.size)} R4 : D $\rightarrow$ D ₁ , id {D.type= D ₁ .type; D.size = D ₁ .size; print(id.name, D.type, D.size)} R5 : T $\rightarrow$ int {T.type = int; T.size = 4} R6 : T $\rightarrow$ float {T.type = float; T.size = 4}	R1 : dlist $\rightarrow$ dlist D { } R2 : dlist $\rightarrow$ D { } R3 : D $\rightarrow$ T L ; { for each x in L.list print(x, T.type) } R4: L $\rightarrow$ L ₁ , id {L.list = append(L ₁ .list, id.name)} R5 : L $\rightarrow$ id { L.list = createlist(id.name)} R6 : T $\rightarrow$ int {T.type = int; T.size = 4} R7 : T $\rightarrow$ float {T.type = float; T.size = 4}
Attributes : id.name, T.type, T.size, D.type, D.size Functions : print()	Attributes : id.name, T.type, T.size, L.list Functions : print(), createlist(), append()

General Structure of Symbol Table

The first entry of the table is reserved to describe basic information about the table itself, such as “global table”, “table for main()”, the number of valid declarations, and the total space in bytes for all the symbols in the scope.

- A filled symbol table for a sample program is shown in the following table.
- The first row of the table gives information about the symbol table itself. For instance, the 0th entry states that this is the global symbol table, it requires 13 bytes of space and has 4 declarations in it.
- The other rows of the table give information about each declaration in the order it is encountered in the file.

Program	index	Name	Type	Size	declarations / offset
	[0]	Global	symbol table	13	4
int a, b;	[1]	a	int	4	0
char ch1;	[2]	b	int	4	4
float f1;	[3]	ch1	char	1	8
	[4]	f1	float	4	9

The purpose of the offset during compilations is to determine the relative layout in memory for all the declared symbols, for example, location of a is from byte 0 to byte 3, location of b is from byte 4 to byte 7, and so on.

Start byte	0	4	8	9
identifier	a	b	ch1	f1